

Exercise 3: Selective Search for Object Recognition

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Questions

Q5.1.1 Why use Selective Search when Felzenszwalb already generates segmentations?

- Felzenszwalb creates oversegmentation with many small regions
- These regions don't correspond to complete objects
- Selective Search hierarchically groups regions into meaningful object proposals
- Enables object detection by generating multiple scales of proposals

Q5.1.2 What criteria for filtering proposals and their effects?

- **Current criteria:** Size > 2000 pixels, aspect ratio < 1.2
- **Effect:** Reduces 1000 proposals to 20-30 meaningful ones
- **Agreement:** Yes, filters noise while keeping real objects
- **Additional criteria:** Eccentricity, solidity, overlap threshold, confidence scores

Q5.1.3 How to obtain rectangles from arbitrary region shapes?

- Find minimum and maximum x,y coordinates of all pixels in region
- Create axis-aligned bounding box from these coordinates
- Convert irregular shapes into rectangular proposals
- Rectangle defined as (min_x, min_y, max_x, max_y)